▧ ITA Spline (Isosceles Triangle Arc Spline)
A 3D curve interpolation method using base-tangent arcs of isosceles triangles

📝 Overview
The ITA Spline (Isosceles Triangle Arc Spline) is a curve interpolation method that connects N points in 3D space with 2(N−1) circular arcs, ensuring C^1 continuity.

Key ideas:

Each segment is constructed from two circular arcs

Arcs are determined uniquely by a geometric construction using isosceles triangles

Works naturally in 3D, even when auxiliary lines are skew

Curve length can be computed analytically

When points lie on a circle, the method can reproduce the circle exactly (circumcenter mode)

The method was originally developed in the 1990s for 3D CAD applications such as walk-through path generation and hose/pipe modeling using partial tori.
Recent literature searches and AI-assisted verification indicate that the method has not been previously published, so it is documented here as the ITA Spline.

🎯 Background
In early 3D CAD systems, it was necessary to:

Generate smooth walk-through paths

Compute walking time accurately

Model hoses and pipes using partial torus primitives

To meet these requirements using only circular arcs, the ITA Spline was devised.
It provides smooth C^1 curves while remaining simple, robust, and computationally lightweight.

✨ Features
✓ 1. C^1 curve composed entirely of circular arcs
Each interval between points is represented by two circular arcs, connected with tangent continuity.

✓ 2. Unique solution via geometric construction
Unlike general biarc interpolation (which has one degree of freedom),
the ITA Spline uses a pair of isosceles triangles and their base-tangent arcs,
which eliminates the degree of freedom naturally and yields a unique solution.

✓ 3. Works naturally in 3D
Even when the auxiliary lines are skew in 3D space,
the isosceles triangles can still be constructed, and the arcs connect smoothly.

✓ 4. Exact circle reproduction (circumcenter mode)
If the input points lie on a circle, the ITA Spline with circumcenter-based direction vectors reproduces the circle exactly.

✓ 5. Fast and lightweight (O(N))
Each segment is computed in constant time using only 3D vector operations.
No special functions or iterative solvers are required.

✓ 6. Analytical arc length
Since the curve is composed of circular arcs,
the total length can be computed exactly.

🛠 Usage (C++ Implementation)
1. Generate arc segments from point list
```
int ITA_Points_to_Arcs(vector<Vct3> vecPt, int Type, vector<strct_arc>& vecArc);
```
2. Compute total arc length

double ArcLength(vector<strct_arc> vecArc);
3. Evaluate a point on the curve (parametric use)
int Point_in_Arcs(vector<strct_arc> vecArc, double s, Vct3& Pt, double& radius);
This function allows the ITA Spline to be used as a parametric curve,
useful for animation, motion paths, and simulation.

📌 Examples
3-point interpolation (open / closed)

Regular pentagon on a circle (exact circle reproduction)

Tornado-shaped 3D curve (0.02% length error vs. analytic solution)

(2,3) torus knot interpolation

📄 License
MIT License

🙏 Acknowledgements
Literature search and originality verification were assisted by Microsoft Copilot.